CIE-I Scheme & Solution.

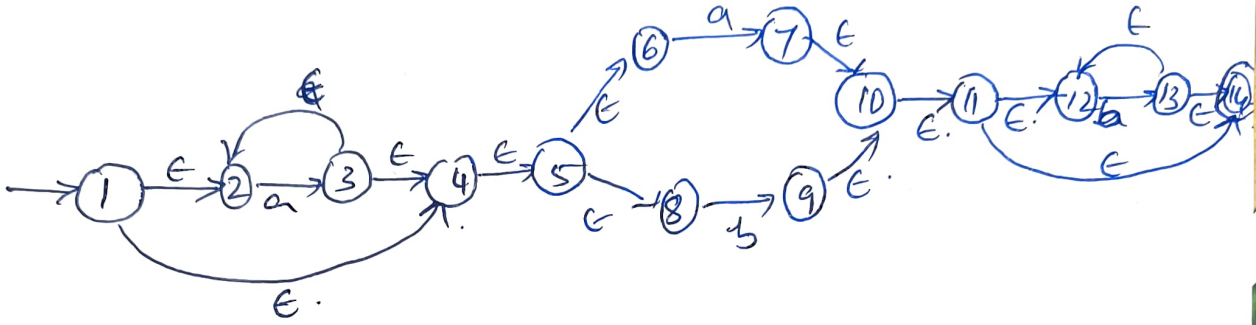
1.1 1

1.2 11

1.3 $aa(a+b)^*aa + ab(a+b)^*ab + ba(a+b)^*ba + bb(a+b)^*ba$.

1.4 ab, bb, aba, bba.

1.5 $a^*(a+b)b^*$: 2M.



1.6 $h(R) = (ab+bc)^*ababbc(ab+bc)^*$.

1.7 Extended transition function for DFA

$\delta^* : Q \times \Sigma^* \to Q$.

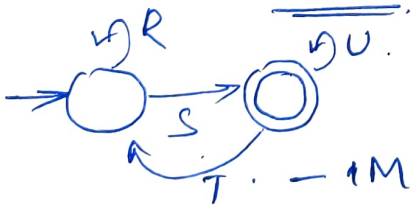
a. for every $q \in Q$, $\delta^*(q, \epsilon) = q$.

b. For every $q \in Q$, $y \in \Sigma^*$, every $\sigma \in \Sigma$.

$$\delta^*(q, y\sigma) = \delta(\delta^*(q,y), \sigma).$$

PART-B

2a)

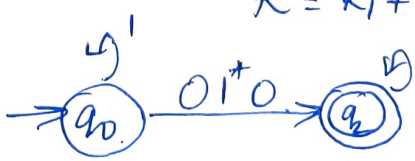


— 1M.

$(R+SU^*T)^*SU^*$. 3 regular expressions

$R = R_1 + R_2 + R_3$.

R_1 Eliminate q_1.



$R_1 = (1+\epsilon+\phi)^*01^*0(0+11^*0)^* = 1^*01^*0(0+11^*0)^*$

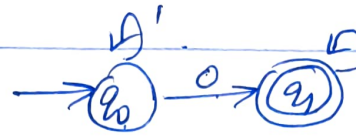
— 2M.

R_2 Eliminate q_2.

$R_2 = 1^*$.

— 1M.

Eliminate. q_2. (R_3).



$R_3 = 1^*0(1+00^*1)^*$.

$R = 1^*01^*0\,(0+11^*0)^* + 1^* + 1^*0\,(1+00^*1)^*$.

– 2M.

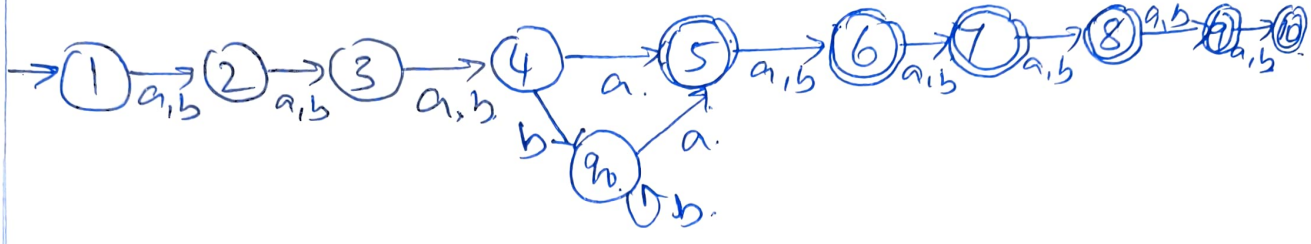
2b) i) Consider regular expression r.
Convert r to ε-NFA using theorem.
Convert ε-NFA to DFA. M1.
Minimize M1
Minimize M. – 2M
Compare M and M1 to decide whether DFA M & regular expression r accepts the same language.
ii) Similary for two ε-NFA's. – 2M.

3a) ε-close (1) = {1,2,5,6}
ε-close (2) = {2}
ε-close (3) = {3}
ε-close (4) = {4}.
ε-close (5) = {5,6}
ε-close (6) = {6}.
ε-close (7) = {7}.

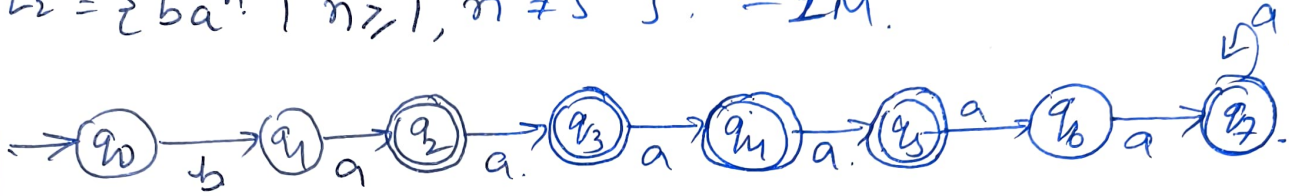
a) Finding ε-closure – 2M.
b) Converting ε-NFA to NFA – 4M.

State	a	b
→*1	{1,2,3,5,6}	{5,6,7}
*2	{3}	ϕ
3	ϕ	{4}
4	ϕ	{2}
*5	ϕ	{5,6,7}
*6	ϕ	{7}
7	{6}	ϕ

3b) $L_1 = w_1 a w_2 : |w_1| \geq 3, |w_2| \leq 5$ — 3M



$L_2 = \{ba^n \mid n \geq 1, n \neq 5\}$. — 1M



4a) Applications with an example. — 3M

i) Lexical analysis.

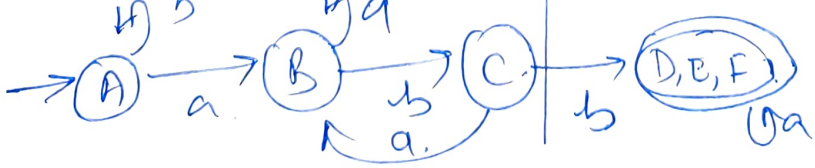
ii) Text search & Pattern Matching.

iii) Unix regular expression.

4b) DFA.

	a	b
→ (A) q_0	$\{q_0, q_1\}$ B	q_0 A
(B) $\{q_0, q_1\}$	$\{q_0, q_1\}$ B	$\{q_0, q_2\}$ C
(C) $\{q_0, q_2\}$	$\{q_0, q_1\}$ B	$\{q_0, q_3\}$ D
* (D) $\{q_0, q_3\}$	q_0, q_1, q_3 E	$\{q_0, q_3\}$ D
* (E) $\{q_0, q_1, q_3\}$ E	$\{q_0, q_1, q_3\}$ E	$\{q_0, q_2, q_3\}$ F
* (F) $\{q_0, q_2, q_3\}$	$\{q_0, q_1, q_3\}$ E	$\{q_0, q_3\}$ D

B	X				
C	X	X			
D	X	X	X		
E	X	X	X	=	
F	X	X	X	=	=
	A	B	C	D	E



5 (i). $L_1 \cup L_2$ — 2M.

Initial state $\{A, x\}$.

Final states $\{C, x\}, \{C, y\}, \{C, z\}$.
$\{A, 2\}, \{B, z\}$.

(ii) $L_1 \cap L_3$ — 2M

Initial state. $\{A, x\}$

Final state $\{C, 1\}, \{C, 2\}$.

(iii). L_3^R — 2M

Introduce new initial state P_0 with ϵ transitions to 1 and 2. and reverse all the transitions.

(iv) $L_2 - L_3$ — 2M

Initial state. $\{x, 1\}$.

Final state. $\{z, A\}, \{z, B\}$.

(v) $\overline{L_2}$ — 2M.

- Swap initial & final states.

6a) Pumping lemma statement — 1M.
proof. — 3M.
$L = \{a^n b^{2n} \mid n \geq 1\}$ proving — 3M.

6b) i. aab.
ii. abab.
iii. bba.
iv. abba.